CISS240: Introduction to Programming
Quiz q1103

Name: eagraham1@cougars.ccis.edu Score:

This is a closed-book, no compiler, 5 minute quiz.

Q1. Write a program that prints solutions to the $x^3 + 3y^3 = nz^3$ where n is an integer input from the user and x, y, z are integers in the interval $[-10, 10]$. After the solutions are printed, the number of solutions is printed. Here's a test case:

```
5
-10 5 -5
-8 4 -4
-6 3 -3
-4 2 -2
-2 1 -1
0 0 0
2 -1 1
4 -2 2
6 -3 3
8 -4 4
10 -5 5
11
```

Answer:

```
#include <iostream>
#include <cmath>

int main()
{
    int n;
    std::cin >> n;

    int count = 0;

    for (int x = -10; x <= 10; ++x)
    {
        for (int y = -10; y <= 10; ++y)
        {
            for (int z = -10; z <= 10; ++z)
            {
                if (pow(x, 3) + 3 * pow(y, 3) == n * pow(z, 3))
                {
                    std::cout << x << " " << y << " " << z << '\n';
```

```
                    count++;
                }
            }
        }
    }

    std::cout << count << '\n';

    return 0;
}
```

Q2. Write an ASCII art program that does the following. 3 test cases are shown.

```
3
  *
 *
*
```

```
4
   *
  *
 *
*
```

```
5
    *
   *
  *
 *
*
```

Answer:

```
#include <iostream>

int main()
{
    int n;
    std::cin >> n;

    for (int i = 0; i < n; ++i)
    {
        for (int j = 0; j < n - i - 1; ++j)
        {
            std::cout << " ";
        }
        std::cout << "*";
```

```
        std::cout << '\n';
    }

    return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.